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2D Finite Element Analysis of a Microstrip Transmission Line

MTH321 Project 2

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Abstract

This report studies the two-dimensional quasistatic capacitance problem associated with a finite-thickness microstrip transmission line. The conducting trace is removed from the finite-element field domain and imposed as an internal Dirichlet boundary; the ground plane is held at zero potential, and the artificial top and side boundaries use homogeneous Neumann conditions. A conforming triangular P_1 finite-element method is verified on a manufactured Poisson problem before being applied to the microstrip geometry. The numerical evidence chain includes baseline matrix and physics checks, a controlled far-boundary truncation study, an energy-norm uniform-refinement study against an over-refined reference, and a deterministic graded-refinement comparison near the conductor corners. A residual-driven adaptive extension was then tested using a flux-jump indicator and Dörfler marking. At a relative energy error of 0.30, it reduced the required DOFs and matrix nonzeros relative to uniform refinement, although its cumulative algorithm time was higher. The final dielectric capacitance is selected from a nested uniform M3–M7 sequence. On the frozen M7 mesh, the computed capacitance is $C = 120.620\,\mathrm{pF/m}$; the matched all-air solve gives $C_{\mathrm{air}} = 35.9682\,\mathrm{pF/m}$ and $\varepsilon_{\mathrm{eff}} = 3.3535$. The resulting quasistatic characteristic impedance is $Z_0 = 50.642\,\Omega$. Hammerstad–Jensen and IPC-2141 engineering approximations give $50.201\,\Omega$ and $49.393\,\Omega$, respectively, agreeing within a few percent. Benchmark agreement is treated as an engineering cross-check, while the primary numerical validation comes from verification, mesh convergence, and matched-mesh consistency tests.

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1 Introduction

A microstrip transmission line consists of a conducting signal trace above a conducting reference plane, separated by a dielectric substrate and surrounded by air. When the electrical length is not negligible compared with the relevant signal wavelengths, the characteristic impedance of the line controls reflection behaviour and therefore signal integrity. Under a lossless quasistatic or quasi-TEM model, this impedance can be inferred from the capacitance per unit length of the cross-section and the corresponding all-air capacitance [3, 6]. The report structure follows the project requirement that numerical claims be supported by verification, convergence evidence, cost information, and reproducible artefacts [4, 5].

The purpose of this project is not merely to produce one number close to $50\,\Omega$. The scientific question is whether a finite-element calculation of the microstrip capacitance can be trusted. The study therefore follows an evidence chain: first a known-solution solver verification, then physical and algebraic checks on the microstrip solve, domain-truncation control, mesh-convergence evidence in an energy norm, a corner-focused graded-refinement experiment, a residual-driven adaptive refinement extension, final capacitance selection, a matched all-air solve, and finally independent engineering formula comparisons.

The nominal geometry is a copper trace of width $w = 3.0\,\text{mm}$ and thickness $t = 0.035\,\text{mm}$ on a substrate of height $H = 1.6\,\text{mm}$ and relative permittivity $\varepsilon_r = 4.4$. All computations use SI units internally. The finite computational domain has $x \in [-W, W]$ and $y \in [0, H_{\text{air}}]$; the final engineering domain uses $W/H = H_{\text{air}}/H = 8$ after a separate domain study.

2 Physical and Mathematical Model

2.1 Geometry and material model

The substrate occupies $0 < y < H$ and has $\varepsilon_r = 4.4$. The air region is above the substrate and has $\varepsilon_r = 1$. The signal conductor is centred at $x = 0$ and occupies $H \le y \le H + t$ and $-w/2 \le x \le w/2$. Its interior is removed from the field domain, so its boundary is an internal Dirichlet boundary rather than a material subregion.

Table 1: Physical and computational parameters for the final microstrip model.

Quantity	Value	Meaning
H	$1.6 \times 10^{-3}\,\text{m}$	substrate height
w	$3.0 \times 10^{-3}\,\text{m}$	trace width
t	$3.5 \times 10^{-5}\,\text{m}$	trace thickness
ε_r	4.4	substrate relative permittivity
V_0	1 V	trace potential
W/H	8	final half-width ratio
H_{air}/H	8	final height ratio

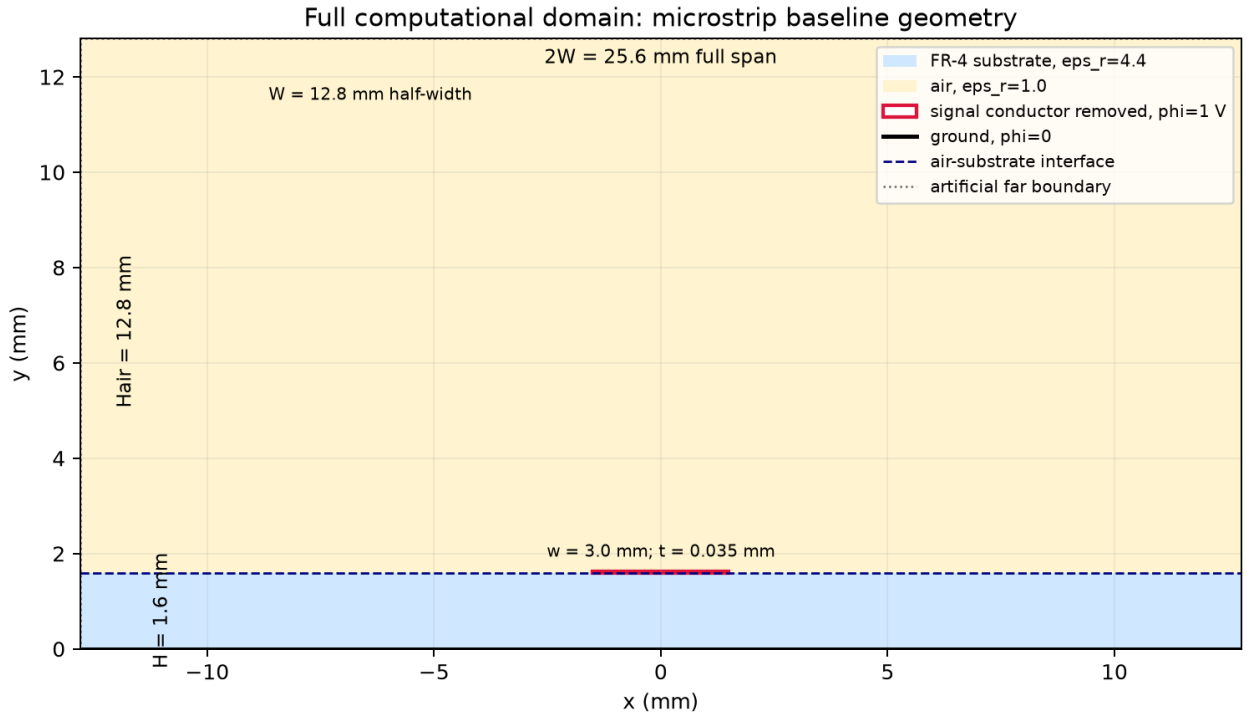


Figure 1: Final geometry convention. The domain half-width is W, so the full horizontal span is $2W$. The trace interior is removed from the field domain and the outer top and side boundaries are artificial truncation boundaries.

2.2 Governing equation and boundary conditions

The potential ϕ satisfies

$$-\nabla \cdot (\varepsilon \nabla \phi) = 0 \qquad \text{in } \Omega, \tag{1}$$

where $\varepsilon = \varepsilon_0 \varepsilon_r$ is piecewise constant. The essential boundary conditions are

$$\phi = V_0 \quad \text{on the signal trace,} \qquad \phi = 0 \quad \text{on the ground plane.}$$

The artificial top and side boundaries use the homogeneous Neumann condition

$$\varepsilon \nabla \phi \cdot n = 0.$$

This same far-boundary convention is used throughout the domain-truncation sequence. Across the air–substrate interface, the exact electrostatic interface conditions are continuity of potential and balance of normal displacement flux.

3 Finite Element Formulation

Let Γ_D denote the union of the trace and ground Dirichlet boundaries. The affine trial space is

$$V_g = \{u \in H^1(\Omega) : u = V_0 \text{ on the trace, } u = 0 \text{ on the ground}\},$$

and the homogeneous test space is

$$V_0 = \{v \in H^1(\Omega) : v = 0 \text{ on } \Gamma_D\}.$$

Multiplying (1) by $v \in V_0$ and integrating by parts gives

$$\int_\Omega \varepsilon \nabla \phi \cdot \nabla v \, d\Omega - \int_{\partial\Omega} \varepsilon \nabla \phi \cdot n \, v \, ds = 0.$$

The Dirichlet part of the boundary term vanishes because $v = 0$ there, and the artificial Neumann term is zero by construction. The weak problem is therefore:

$$\text{find } \phi \in V_g \text{ such that } \int_\Omega \varepsilon \nabla\phi \cdot \nabla v \, d\Omega = 0 \quad \forall v \in V_0.$$

The discrete space uses conforming triangular P_1 elements. Writing

$$\phi_h = \sum_j \Phi_j N_j,$$

the stiffness matrix has entries

$$K_{ij} = \int_\Omega \varepsilon \nabla N_i \cdot \nabla N_j \, d\Omega.$$

Because ε is discontinuous at the material interface, the integral is evaluated element by element using each triangle's material tag. In a conforming H^1 finite-element space, ϕ_h is continuous across the air–substrate interface. Since the discontinuous coefficient is integrated in the same global weak form, the interface flux-balance condition is enforced automatically in the weak or distributional sense. No separate Lagrange multiplier is required. This does not mean that the two elementwise P_1 normal-flux traces $\varepsilon\nabla\phi_h \cdot n$ are pointwise identical on every interface edge.

The implementation uses scikit-fem version 12.0.2 for the P_1 basis functions, quadrature, and sparse finite-element assembly. The project code explicitly defines the geometry, trace hole, material subdomains, boundary tags, refinement studies, capacitance extraction, and validation logic. The library provides reliable assembly tools, but it does not by itself validate the physical model.

Dirichlet conditions are imposed by eliminating the trace and ground degrees of freedom, leaving a sparse symmetric positive definite free-degree system solved by a sparse direct solver.

The capacitance is extracted in two ways. The energy method is

$$W_e = \frac{1}{2}\int_\Omega \varepsilon |\nabla\phi_h|^2 \, d\Omega, \qquad C_E = \frac{2W_e}{V_0^2}.$$

The reaction-charge method sums the assembled Dirichlet reactions on the trace:

$$C_Q = \frac{|Q_{\text{trace}}|}{|V_0|}.$$

Agreement between C_E and C_Q verifies the sign convention, Dirichlet dof selection, and quadrature. It is an internal consistency check, not a spatial-convergence proof.

4 Solver Verification and Baseline Checks

4.1 Manufactured-solution verification

Before using the microstrip result, the implementation was verified on a smooth known solution. The test problem uses a unit square with zero Dirichlet data and exact solution

$$u(x, y) = \sin(\pi x)\sin(\pi y).$$

The manufactured forcing and boundary data are explicitly

$$-\Delta u = f, \qquad f(x, y) = 2\pi^2 \sin(\pi x)\sin(\pi y), \qquad u|_{\partial(0,1)^2} = 0.$$

Here the named errors are $\|u - u_h\|_{L^2(\Omega)}$ and $\|u - u_h\|_E = (\int_\Omega |\nabla(u - u_h)|^2 \, d\Omega)^{1/2}$, with coefficient one for this verification problem. The expected smooth P_1 rates are two and one, respectively. The five consecutive meshes below contain the required $h, h/2, h/4, h/8$ protocol and one further level $h/16$. Pairwise orders are $p_i = \log(E_{i-1}/E_i)/\log(h_{i-1}/h_i)$.

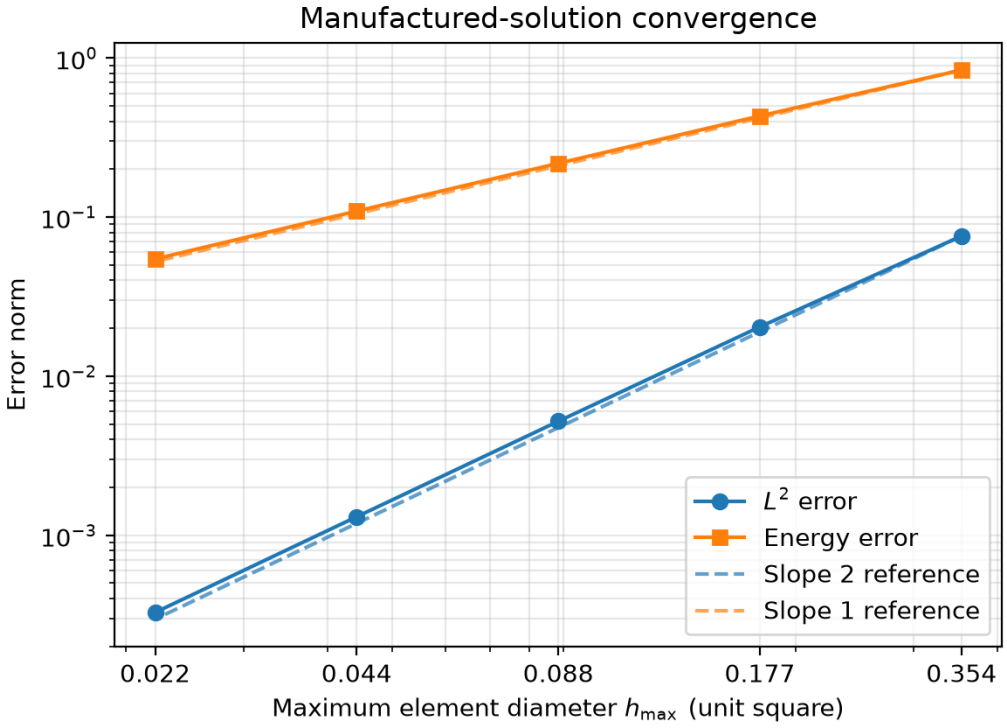


Figure 2: Manufactured-solution convergence. The finest-step observed orders are approximately 2.00 in L^2 and 1.00 in the energy norm, as expected for smooth P_1 finite elements.

Table 2: Consecutive manufactured-solution refinements. N_T is the triangle count and r_F is the free-system algebraic residual. All five levels use a sparse direct solve; an iterative stopping tolerance is not applicable at any level.

Level	N_T	DOFs	$h_{\max}$	$\|e\|_{L^2}$	$\|e\|_E$	$\|r_F\|_\infty$	p_{L^2}	p_E
2 (h)	32	25	0.35355	7.596×10^{-2}	8.397×10^{-1}	3.331×10^{-16}	–	–
3 ($h/2$)	128	81	0.17678	2.041×10^{-2}	4.320×10^{-1}	5.551×10^{-16}	1.896	0.959
4 ($h/4$)	512	289	0.08839	5.201×10^{-3}	2.176×10^{-1}	9.021×10^{-16}	1.972	0.989
5 ($h/8$)	2048	1089	0.04419	1.307×10^{-3}	1.090×10^{-1}	1.457×10^{-15}	1.993	0.997
6 ($h/16$)	8192	4225	0.02210	3.270×10^{-4}	5.451×10^{-2}	2.368×10^{-15}	1.998	0.999

Least-squares fits of $\log E$ against $\log h_{\max}$ over the finest three levels (4–6) give $p_{L^2} = 1.9956$ and $p_E = 0.9983$. These fitted orders complement the pairwise orders rather than selecting nonconsecutive meshes. The largest listed free residual is below 2.4×10^{-15}; the verification residual acceptance check is $\|r_F\|_\infty < 10^{-10}$, not an iterative solver tolerance.

4.2 Baseline microstrip sanity checks

The representative baseline solve passed the expected physical and algebraic checks. The potential remained in the interval $[0, 1]$, the symmetry error about $x = 0$ was near round-off, the reduced matrix symmetry error was zero to numerical precision, and the residual was negligible. The energy and reaction capacitances agreed to within 2.54×10^{-15} relative difference.

Linear solver and algebraic-error control. After Dirichlet elimination, the free system is sparse, symmetric and positive definite, so a sparse direct solve is appropriate for the tested problem sizes. This choice also avoids ambiguity about iterative stopping tolerances. A small residual alone would not prove negligible algebraic error, so the residual is interpreted together with conditioning diagnostics. For the representative baseline solve, the estimated physical reduced-system condition number was

1.7922363×10^5, while the diagonal-scaled diagnostic was 1.4249857×10^4. The free residual infinity norm was 3.101927×10^{-25}. The scaled diagnostic is not the physical condition number, but these conditioning and backward-residual diagnostics indicate that algebraic solve error is far below the observed finite-element discretisation effects.

Table 3: Representative baseline checks on the $8H$ microstrip domain.

Diagnostic	Value
Potential range	$[0, 1]$
Symmetry relative error	3.244×10^{-14}
Reduced-matrix symmetry error	0
Smallest eigenvalue estimate	5.918×10^{-15}
Largest eigenvalue estimate	1.060677×10^{-9}
Condition estimate	1.7922363×10^5
Diagonal-scaled condition diagnostic	1.4249857×10^4
Free residual infinity norm	3.101927×10^{-25}
C_E and C_Q diagnostic difference	2.536×10^{-15}
Charge-balance defect	6.973×10^{-15}

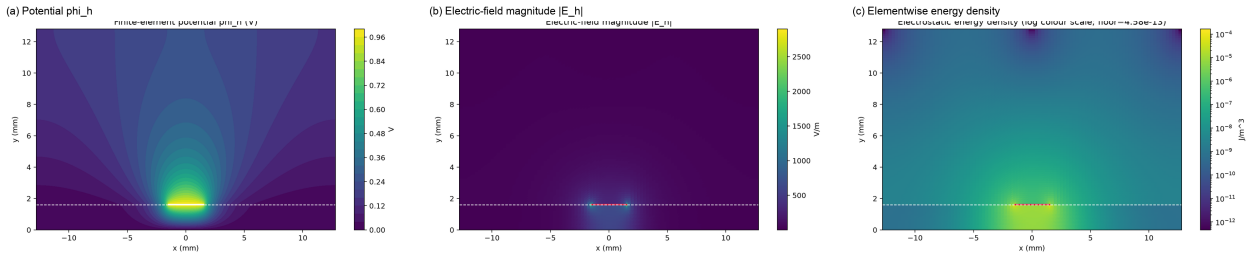


Figure 3: Baseline field diagnostics: (a) finite-element potential ϕ_h, (b) electric-field magnitude $|E_h|$, and (c) elementwise electrostatic energy density, shown with a logarithmic colour scale for visibility. The potential respects the imposed Dirichlet values, while the field and energy density concentrate near the trace corners. These plots are diagnostic visualisations, not error norms.

5 Domain Truncation

The physical air region is unbounded, but the computation uses a finite rectangle. To isolate the truncation effect, the domain study used the sequence $(W/H, H_{\text{air}}/H) = (3, 3), (5, 5), (8, 8), (12, 12)$ while preserving the near-trace mesh. The $12H$ result was used only as a tested finite-domain reference, not as the exact infinite-domain solution.

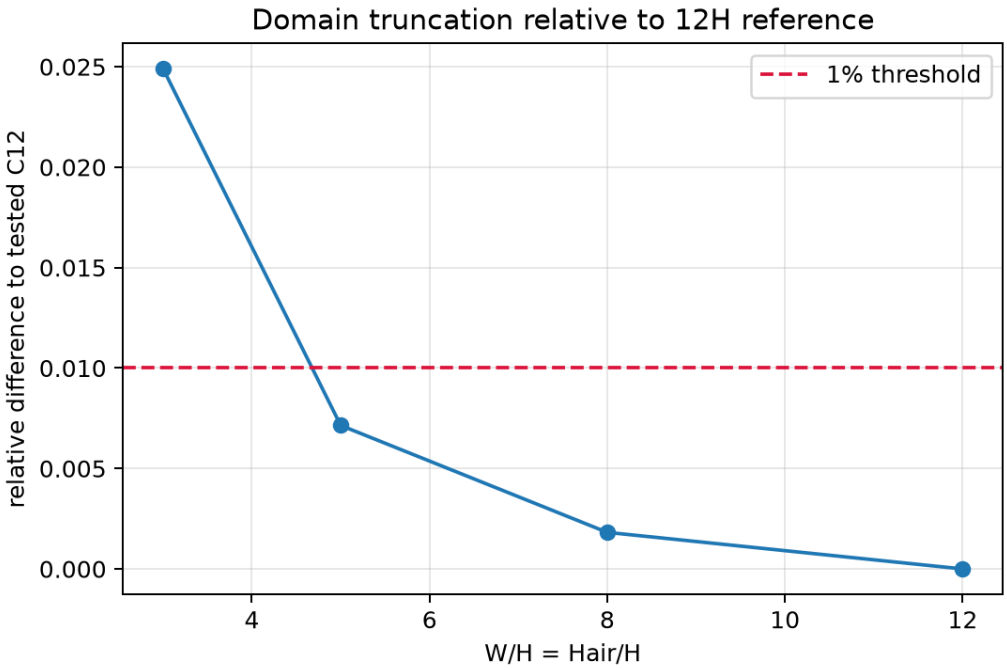


Figure 4: Domain-truncation study. Differences are relative to the tested $12H$ finite-domain reference. The horizontal line marks the one-percent threshold.

Table 4: Production domain-truncation results.

$W/H = H_{\text{air}}/H$	Elements	DOFs	C_E (pF/m)	Difference from $12H$
3	11868	6121	118.839	2.491%
5	16988	8711	121.004	0.714%
8	26168	13346	121.653	0.182%
12	41208	20926	121.875	reference

The largest-domain mesh-adequacy check repeated the $12H$ case with a finer near-field resolution. The production value was $C_{12,R0} = 1.2187452011154754 \times 10^{-10}\,\text{F/m}$, while the refined repeat gave $C_{12,R1} = 1.2183061087416544 \times 10^{-10}\,\text{F/m}$. The resulting mesh sensitivity was 3.604122×10^{-4}, or about 0.0360%. This is smaller than the $8H$–to–$12H$ production difference of about 0.182%, so the selected-domain conclusion is not driven primarily by the production near-field mesh. Neither $12H$ value is treated as the exact infinite-domain capacitance.

The smallest tested domain below one percent was $5H$. The later studies use the $8H$ domain because its tested difference from $12H$ was only about 0.182% while remaining substantially cheaper than the largest domain. This is a finite-domain sensitivity indicator rather than a rigorous infinite-domain error bound.

6 Mesh Convergence and Corner Singularity

6.1 Corner-field behaviour

The conductor corners create singular electric-field behaviour. The lower trace corners are especially delicate because the conductor boundary meets the air–substrate interface. A mesh-dependent peak field that increases under refinement is therefore consistent with singular behaviour; it is not itself an accuracy norm.

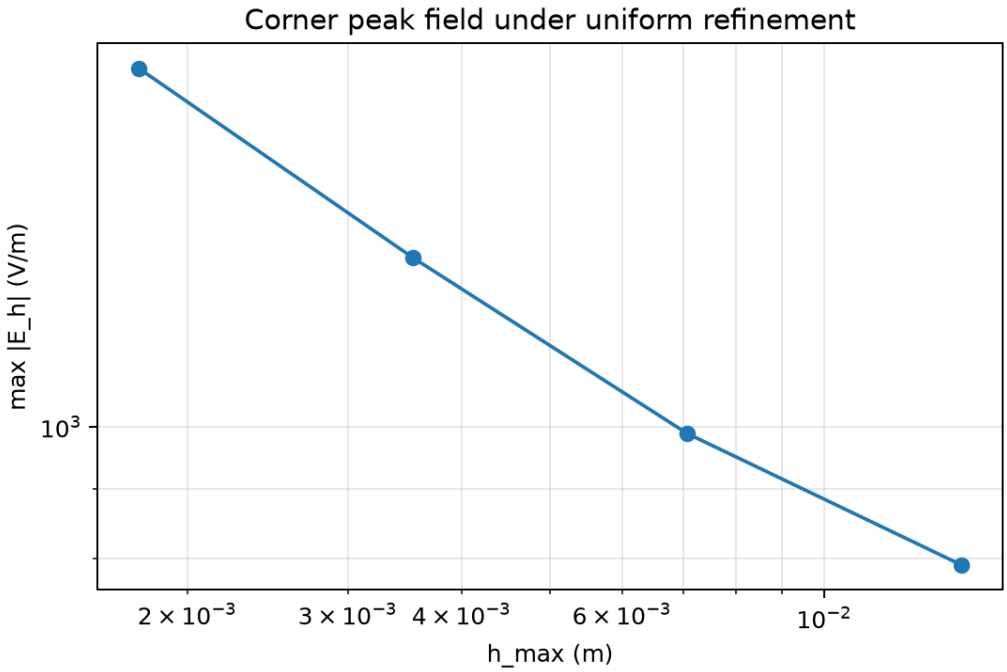


Figure 5: Maximum computed electric-field magnitude under uniform refinement. The increasing peak is consistent with stronger resolution of the trace-corner field.

6.2 Uniform energy-norm convergence

Uniform convergence was measured against an over-refined nested reference $R2 = M6$. The error norm was

$$\|e\|_E^2 = \int_\Omega \varepsilon |\nabla(\phi_{\text{ref}} - \phi_h)|^2 \, d\Omega.$$

The relative energy error used in the figures and tables is

$$E_{\text{rel}}^{\text{ref}} = \frac{\|\phi_{\text{ref}} - \phi_h\|_E}{\|\phi_{\text{ref}}\|_E}, \qquad \|\phi_{\text{ref}}\|_E^2 = \int_\Omega \varepsilon |\nabla \phi_{\text{ref}}|^2 \, d\Omega. \tag{2}$$

When the reference changes, both numerator and denominator use that same reference; this dimensionless ratio is not a relative capacitance difference. Coefficient vectors on different meshes were never directly subtracted. Instead, the coarse P_1 gradient was evaluated on the reference mesh elements and the error was integrated on the reference grid.

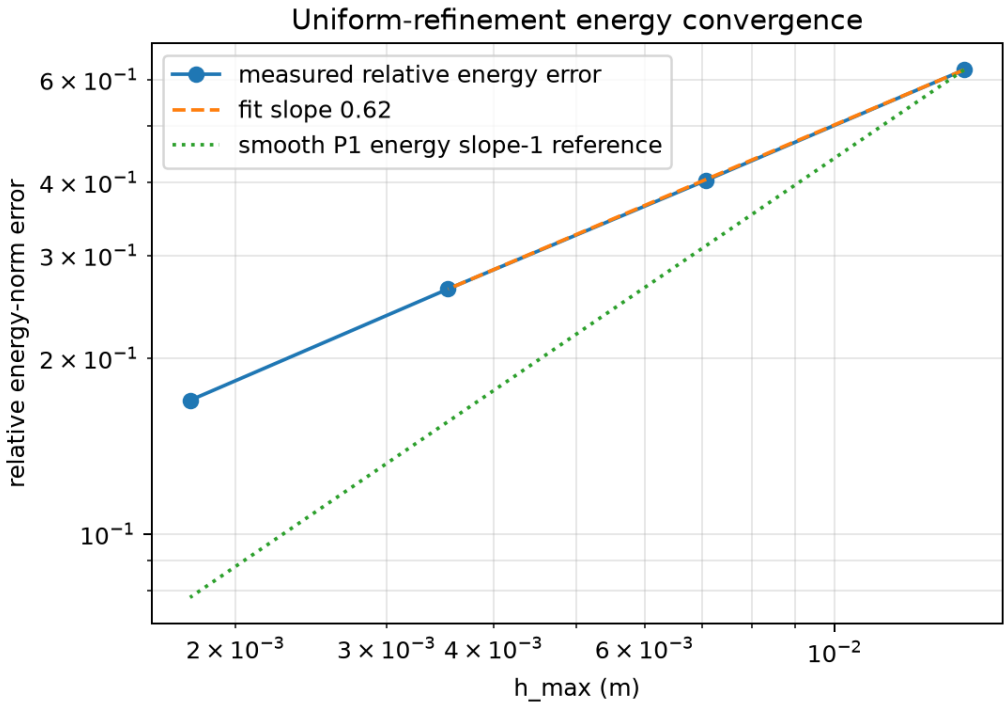


Figure 6: Uniform-refinement energy-norm convergence. The fitted slope over M0–M2 is about 0.623, below the smooth P_1 energy slope of one.

Table 5: Uniform energy-error convergence against the accepted $R2 = M6$ reference.

Mesh	DOFs	$h_{\max}$ (m)	Relative energy error	C_E (pF/m)
M0	35	1.415×10^{-2}	0.6230	167.703
M1	114	7.077×10^{-3}	0.4030	140.432
M2	404	3.538×10^{-3}	0.2627	129.144
M3	1512	1.769×10^{-3}	0.1694	124.273

Table 6: Reference stability in the M0–M2 fit window. R0=M4, R1=M5 and R2=M6. The last column is $|E_{\mathrm{rel}}^{R2} - E_{\mathrm{rel}}^{R1}|/|E_{\mathrm{rel}}^{R2}|$; each relative error is normalised as in (2).

Mesh	E_{rel}^{R0}	E_{rel}^{R1}	E_{rel}^{R2}	$R1 \to R2$ change
M0	0.610595	0.619193	0.623045	0.618%
M1	0.386757	0.398039	0.403041	1.241%
M2	0.239121	0.255580	0.262690	2.707%

The accepted fitted slope is 0.623 over M0–M2. The R1-to-R2 changes in Table 6 range from 0.618% to 2.707%; changing the reference changes the fitted slope from 0.6383 to 0.6230. Thus the qualitative sublinear-rate conclusion survives reference refinement. This finite sensitivity is disclosed rather than called zero or a rigorous negligible-error bound. M3 is excluded from the fit because its R1-to-R2 error sensitivity is larger (6.384%). The reduced rate is consistent with conductor-corner and material-interface regularity loss.

7 Graded Versus Uniform Refinement

A deterministic graded strategy was tested to concentrate coordinate-line refinement near the four trace corners, with additional emphasis on the lower conductor/material-junction corners. The method remained symmetric about $x = 0$ and used the same physical model as the uniform sequence.

Reproducible grading rule. The shared base coordinate sets, in units of H, are

$$B_x/H = (-8, -2, -0.9375, 0, 0.9375, 2, 8), \qquad B_y/H = (0, 1, 1.021875, 1.5, 8).$$

Let $D_k(B)$ bisect every interval of B exactly k times (ratio 1/2). For parameters (b, u, ℓ), retain the background coordinates $D_b(B_x)$ and $D_b(B_y)$. Add coordinates from $D_\ell(B_x)$ within $0.45H$ of either $x = \pm w/2$ and from $D_\ell(B_y)$ within $0.45H$ of $y = H$. Add coordinates from $D_u(B_x)$ within $0.30H$ of either $x = \pm w/2$ and from $D_u(B_y)$ within $0.30H$ of $y = H + t$. Take sorted unions, snap to the M6 coordinate set at 10^{-15} m precision, form the Cartesian grid, and split each retained rectangle along its bottom-left/top-right diagonal; remove the trace interior. These are axis-aligned bands, not circular element-distance zones: adding a coordinate line also spends resolution away from a corner.

Table 7: Deterministic G-level parameters read from the experiment CSV. b is the background depth, u the upper-corner depth, and ℓ the lower-junction depth. The lower corners receive one extra depth for G0–G3; G4 uses depth five for both, with the larger lower-corner band retained.

Mesh	b	u	ℓ	DOFs	Min. corner size (m)
G0	0	1	2	123	4.210×10^{-4}
G1	1	2	3	537	2.105×10^{-4}
G2	1	3	4	1422	1.052×10^{-4}
G3	2	4	5	5306	5.262×10^{-5}
G4	3	5	5	7068	5.262×10^{-5}

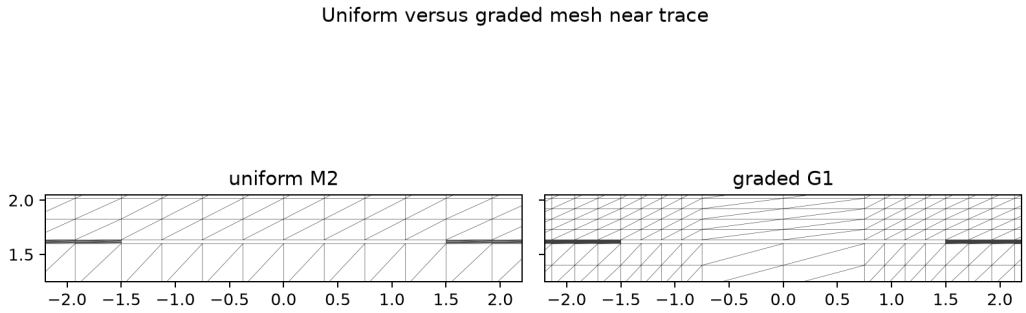


Figure 7: Matched-work visual comparison of uniform and graded meshes. The graded mesh concentrates resolution near the trace corners, whereas the uniform mesh spends resolution throughout the domain.

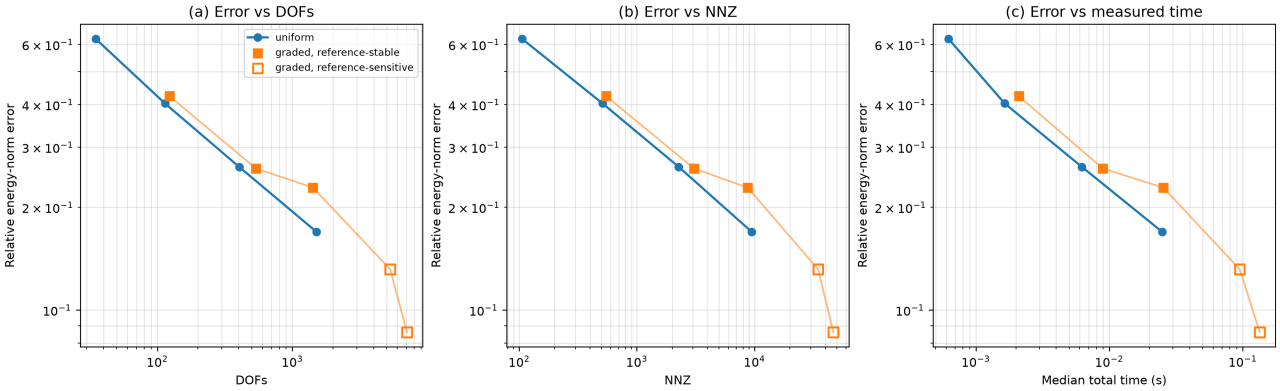


Figure 8: Recorded uniform versus graded relative energy-error estimates against (a) DOFs, (b) NNZ and (c) measured median total time. Filled orange squares meet the reference-sensitivity criterion; open squares do not. A subsequent triangle-geometry audit found an additional graded integration limitation, described in the text; the recorded curves are retained rather than silently recomputed.

The graded sequence improved local corner resolution. In the recorded estimates, targets 0.40 and 0.30 met the reference-sensitivity criterion and gave uniform-to-graded DOF ratios of about 0.798 and 0.783, respectively; NNZ and measured timing gave the same qualitative outcome. The 0.20 target remains exploratory because its interpolation uses a reference-sensitive point. These observations apply to this tested coordinate-line strategy, not all adaptive FEM methods.

Cross-mesh integration audit and remaining limitation. The original graded check verified only that coordinate sets were subsets of R2. That is insufficient for triangular common refinement: when a coarse rectangle has different numbers of subdivisions in x and y, its diagonal can cut reference triangles. A geometry-only audit found such crossings for every G-level; for G2, 21440 of the 180224 R2 triangles are crossed, covering about 4.55% of the field-domain area. This area fraction is not an energy-error bound. The original single-centroid integration is consequently an approximate cross-mesh quadrature, not exact common-refinement integration. The existing numbers are frozen, but precise graded efficiency claims require common-overlay integration or a proven quadrature-convergence check. A corrected guard now rejects this unsupported shortcut before any new solve. The uniform hierarchy used for Section 6 is genuinely nested and does not have this diagonal-crossing defect. The new, separately stored common-overlay comparison in Section 8 supplies that validation without replacing the historical curves above.

This negative result motivated a second refinement strategy in which element selection is driven by the computed flux-jump residual rather than by a prescribed geometric grading pattern.

8 Residual-Based Adaptive Refinement

8.1 Motivation and controlled setup

The reduced uniform energy rate, $p_E \simeq 0.623$, and the absence of a measured advantage for the prescribed coordinate-line grading motivated a solution-driven extension. The hypothesis was recorded before the experiments in `logs/PRE_ADAPTIVE_HYPOTHESIS.md`. No advantage was assumed. The $8H$ domain, trace hole, material coefficient, homogeneous-Neumann far boundary, P_1 formulation and sparse direct solver were held fixed. No new all-air problem was solved, and the final M7 engineering values were not changed.

The historical M2 coordinates are left–right symmetric, but its identically oriented cell diagonals are not. To avoid confusing mesh-orientation asymmetry with adaptivity, the primary initial mesh uses a conforming, mirror-invariant subdivision of M1. All edge midpoints are inserted; each right triangle is divided into four nonobtuse triangles, giving exactly the M2 coordinates, 704 elements and 404 DOFs. We call this *M1-S (M2-resolution)*, not the unchanged M2 triangulation. An eight-cycle

M0-S companion path brackets the 0.40 and 0.30 error targets, since the primary initial error is already below 0.30.

An earlier diagonal-only symmetry completion was rejected: its very coarse M0 companion produced minimum potentials down to about -0.030 V. All its points and timings remain in a separate failed-attempt directory and are excluded from the conclusions. The correction was made to satisfy potential bounds, not to optimise efficiency; the indicator and marking parameter were unchanged.

8.2 Flux-jump indicator and marking

For $f = 0$, constant elementwise ϵ and affine $\phi_h|_K$, the interior volume residual vanishes. For an interior edge, choose one unit normal n_e and define

$$J_e = (\epsilon \nabla \phi_h|_{K^+} - \epsilon \nabla \phi_h|_{K^-}) \cdot n_e.$$

The same normal is used on both sides, including material-interface edges. With h_e equal to edge length, the implemented indicator is

$$\eta_K^2 = \frac{1}{2} \sum_{e \subset \partial K \cap \mathcal{E}_{\text{int}}} h_e \|J_e\|_{L^2(e)}^2 + \sum_{e \subset \partial K \cap \Gamma_{\text{far}}} h_e \|\epsilon \nabla \phi_h \cdot n\|_{L^2(e)}^2, \qquad \eta = \left(\sum_K \eta_K^2 \right)^{1/2}. \tag{3}$$

Since these traces are constant, each term is evaluated exactly as $h_e^2 J_e^2$ (or its boundary counterpart). Trace and ground Dirichlet edges contribute no natural-boundary residual. This is a *residual-based refinement indicator*, not a guaranteed error bound.

At every step, deterministic Dörfler marking chooses the smallest descending-indicator prefix containing at least $\theta = 0.5$ of $\sum_K \eta_K^2$. The reflected triangles are added for symmetry. All edges of marked triangles are split; any triangle with two split edges has its third edge split as well. This conformity closure gives unambiguous red/four-child and green/two-child subdivisions. Minimal marks, symmetry-added marks and all actually refined parents are recorded separately. No corner-distance rule selects elements.

8.3 Reference error and adaptive mesh behaviour

The principal and sensitivity references are M6 and the frozen M7 dielectric solution. For $R \in \{M6, M7\}$,

$$\|e_R\|_E = \left(\int_\Omega \epsilon |\nabla(\phi_R - \phi_h)|^2 \, d\Omega \right)^{1/2}, \quad E_{\text{rel},R} = \frac{\|e_R\|_E}{\|\phi_R\|_E}, \quad S_{\text{ref}} = \frac{|\|e_{M7}\|_E - \|e_{M6}\|_E|}{\|e_{M7}\|_E}.$$

All three methods are newly compared using $E_{\text{rel},M6}$ and the same exact triangle-overlay integration. Contained reference triangles use constant gradients directly; crossed triangles are intersected geometrically and integrated by overlap area. Coefficient vectors on different meshes are never subtracted. This also resolves the integration issue identified in Section 7 for the *new* comparison, without changing its historical Figure 8 or raw tables. The maximum relative overlay area-coverage defect was below 2×10^{-12}. Formal comparisons require $S_{\text{ref}} < 3\%$; other points remain visible with open markers.

Table 8: Primary residual-adaptive sequence. Every point meets the predeclared reference-sensitivity criterion. Full-precision capacitances, marks, sanity checks and timing samples are retained in the extension CSVs.

Cycle	Elements	DOFs	NNZ	E_{rel}	S_{ref} (%)
A0	704	404	2265	0.2637	0.847
A1	758	432	2439	0.2279	1.123
A2	898	504	2891	0.2056	1.325
A3	1218	666	3973	0.1789	1.610
A4	1650	890	5381	0.1670	1.600
A5	2270	1206	7469	0.1553	0.941
A6	3040	1602	10019	0.1434	0.297
A7	4008	2088	13345	0.1354	0.412

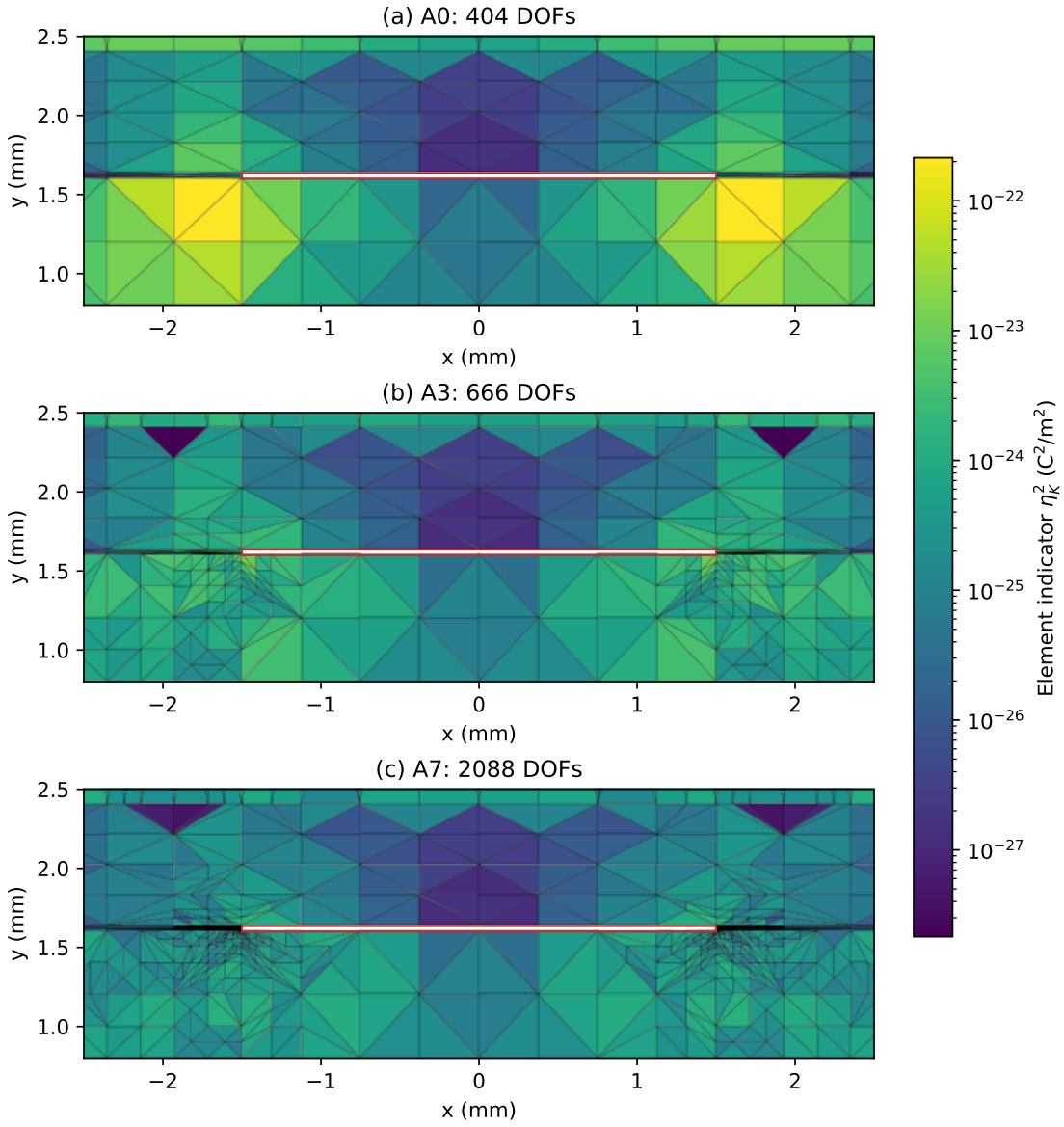


Figure 9: Residual-adaptive meshes A0, A3 and A7 in the same physical near-trace window; colours show η_K^2 on a shared logarithmic scale. Refinement increasingly concentrates around the trace ends, while some marked regions lie farther into the substrate and air. This is observed localisation, not a claim that the indicator exactly identifies every singular contribution.

Within $0.25H$ of a trace corner, the element fraction rises from 5.11% at A0 to 62.0% at A7; these elements contain 79.6% of the final squared indicator. Initially 88.6% of the indicator is assigned to elements whose nearest corner is a lower junction; at A7 this fraction is 50.2%. Thus lower-junction concentration is strongest early, not uniformly dominant throughout. The far-boundary contribution is about 0.00265% of the final indicator. Some final marked element centroids reach $|x| = 5.0$ mm and $y = 4.57$ mm, so refinement is not confined to four prescribed disks.

8.4 Matched-work comparison

One warm-up and five measured repetitions are used for every point. Uniform and graded work includes mesh generation, assembly, solve and capacitance extraction. Adaptive work is cumulative from its initial mesh, including all preceding solves, estimators, marking and refinement, up to the current solution. This accounting deliberately charges adaptive refinement for the preceding solve–estimate–mark–refine cycles because those computations are required to obtain the final adaptive mesh; the comparison is therefore conservative with respect to adaptivity. DOF and NNZ comparisons

remain independent of this timing convention. Offline reference solves and error integration are excluded equally. This prevents comparing an entire adaptive trajectory with only a final sparse solve. Millisecond-scale differences on the coarsest meshes are not interpreted as convincing speedups.

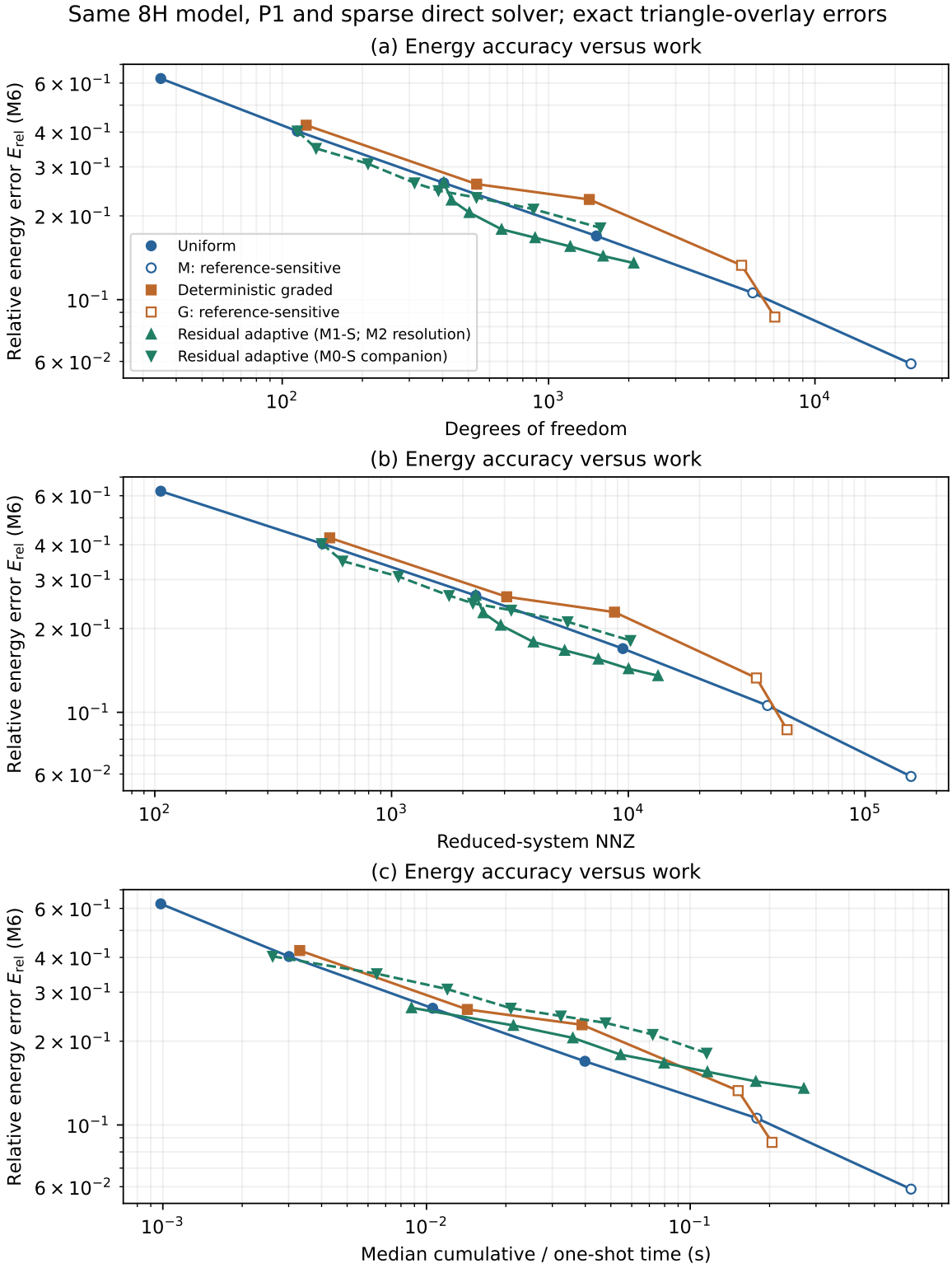


Figure 10: New three-way comparison in a common norm: (a) DOFs, (b) reduced NNZ, (c) measured median total algorithm time. Solid green triangles are the primary M1-S path; dashed green triangles are its separately identified M0-S companion. Open markers fail the M6/M7 reference-sensitivity threshold. Historical uniform/graded raw errors and timing tables were not overwritten.

Table 9: Piecewise log-linear interpolation between measured consecutive points only. U, G, A denote uniform, prescribed grading and residual adaptivity. Ratios above one favour adaptivity. The 0.40/0.30 adaptive targets use the M0-S companion. *The 0.20 three-way comparison is exploratory because the graded bracket includes reference-sensitive G3.

Target	N_U	N_G	N_A	N_U/N_A	$\mathrm{NNZ}_U/\mathrm{NNZ}_A$	T_U/T_A
0.40	116.6	146.2	115.1	1.013	1.022	1.115
0.30	272.8	348.7	224.0	1.218	1.235	0.546
0.20*	917.3	1974.2	532.7	1.722	1.791	0.618

At $E_{\mathrm{rel}} = 0.30$, adaptivity needs about 224 DOFs versus 273 for uniform and 349 for grading, with uniform/adaptive DOF and NNZ ratios 1.218 and 1.235. However, its cumulative median time is 13.1 ms versus 7.15 ms for uniform: the time ratio is 0.546, not a speedup. At 0.40, the DOF ratio is only 1.013 and the measured times are too small to establish a robust runtime advantage. In a direct reference-stable comparison, A5 uses 1206 DOFs with error 0.1553, while M3 uses 1512 DOFs with error 0.1694; A5 nevertheless takes about 2.91 times the cumulative algorithm time.

8.5 Interpretation and capacitance

The extension demonstrates an approximation-size advantage at 0.30, but not a uniform runtime advantage. A7 has 1522 elements outside the specified corner neighbourhoods, versus 2047 for G2 at a comparable scale. This supports the explanation that coordinate-line grading spends more resolution away from corners; it does not isolate the causal effects of band width, lower-corner weighting or transitions, which were not varied independently.

The capacitance diagnostic uses $|C_h - C_{M7}|/|C_{M7}|$, where computed M7 is a selected numerical reference, not exact capacitance. A5 differs by 2.42%, versus 3.03% for M3, so the energy improvement also has a capacitance benefit in this matched-work pair. A7 still differs by 1.74% and is not substituted for the accepted M7 engineering mesh.

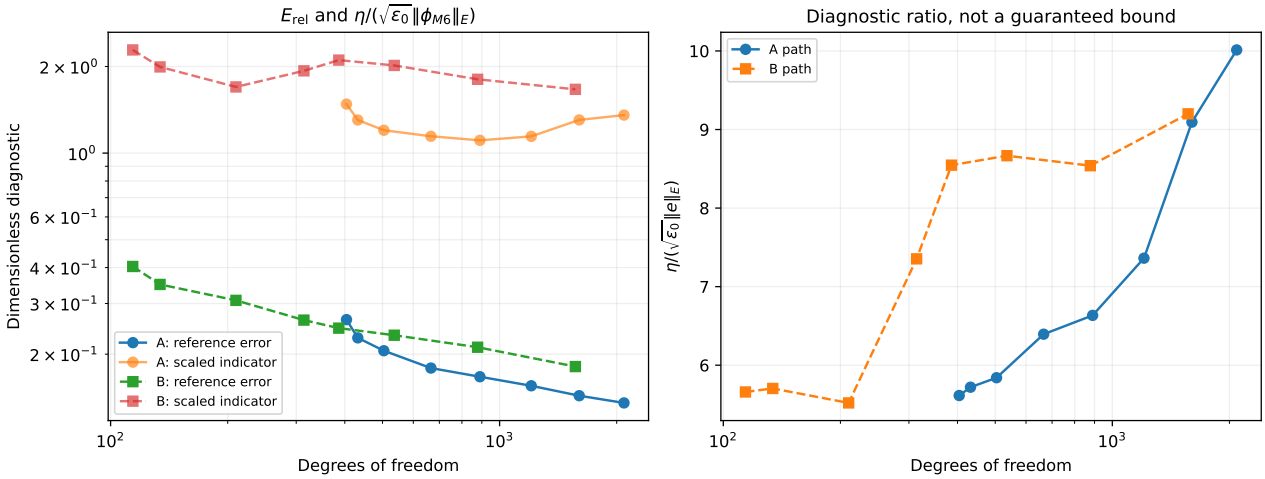


Figure 11: Indicator diagnostics. Physical η has units C/m, whereas $\|e\|_E$ has units $\sqrt{\mathrm{J/m}}$. For a dimensionless display only, η is divided by $\sqrt{\epsilon_0}$ and the reference norm; the requested unscaled ratio $\eta/\|e\|_E$ is also stored. The dimensionless diagnostic ratio increases from about 5.61 to 10.0 on the primary path: it has not stabilised and is not an effectivity constant or guaranteed error bound.

The primary true reference error decreases throughout, but η decreases initially and rises again over the last levels. Therefore this indicator is useful for selecting refinement locations in this experiment, but the data do not justify using it as a calibrated stopping criterion. All corrected A/B cycles satisfy potential bounds, symmetry, residual, matrix-symmetry, charge-balance, material and conformity checks; the failed first attempt remains explicitly archived.

9 Final Capacitance and Effective Permittivity

9.1 Final dielectric capacitance

The final capacitance selection used the nested uniform M3–M7 hierarchy on the fixed $8H$ domain. M8 was preflighted but not recommended because of the expected sparse-direct factorisation cost and fill-in. M7 was accepted through an asymptotic/resource fallback: the capacitance sequence decreased monotonically, the apparent output orders were stable, and the Richardson diagnostic was stable. The final reported capacitance remains the computed M7 FEM value, not the Richardson extrapolate.

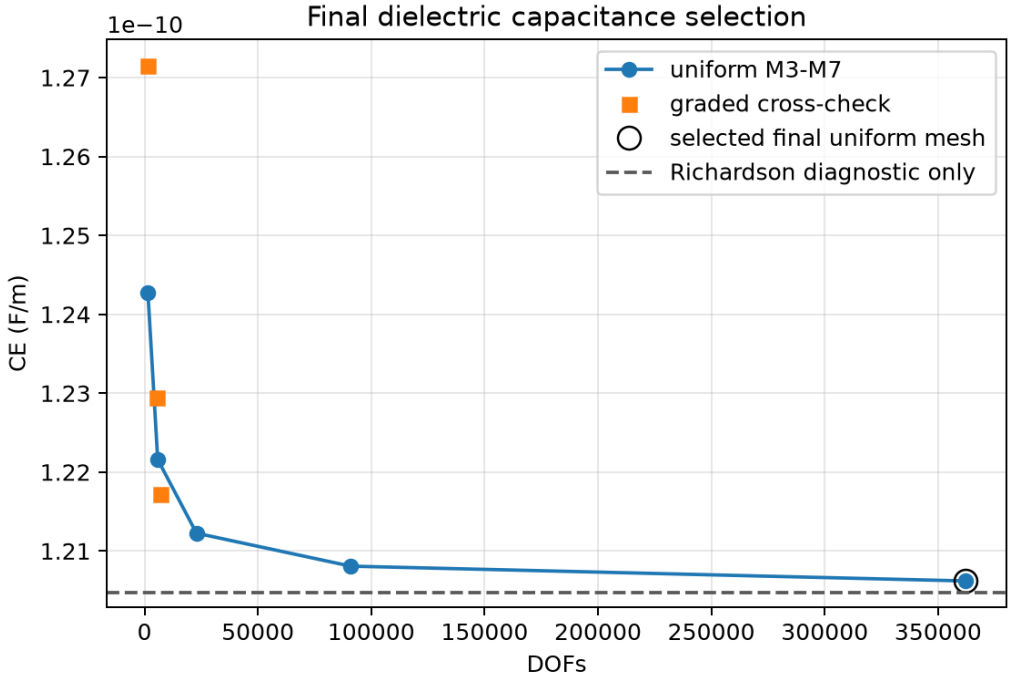


Figure 12: Final dielectric capacitance selection. M7 is the selected computed FEM mesh; the Richardson line is a diagnostic only.

Table 10: Final dielectric capacitance sequence.

Mesh	Elements	DOFs	C_E (pF/m)	Successive change
M3	2816	1512	124.273	–
M4	11264	5840	122.159	1.731%
M5	45056	22944	121.225	0.770%
M6	180224	90944	120.807	0.346%
M7	720896	362112	120.620	0.156%

The M7–M6 change, 0.1556%, is above the preferred direct 0.1% quality-control target. The reported value is the computed M7 result. Its remaining spatial sensitivity is approximately 0.156%, while the tested finite-domain sensitivity is approximately 0.182%; neither is a rigorous error bound. They are reported separately rather than combined statistically.

9.2 Matched all-air solve and effective permittivity

The all-air calculation reused the exact frozen M7 mesh. Only the material coefficient was changed to $\varepsilon_r = 1$ everywhere. This matching is important because $\varepsilon_{\text{eff}} = C/C_{\text{air}}$ is a ratio; changing the mesh would mix material and discretisation effects. The mesh fingerprint, degrees of freedom, boundary sets, and matrix sparsity pattern were all checked to match.

$$C = 120.620\,\mathrm{pF/m}, \qquad C_{\mathrm{air}} = 35.9682\,\mathrm{pF/m}, \qquad \varepsilon_{\mathrm{eff}} = \frac{C}{C_{\mathrm{air}}} = 3.3535.$$

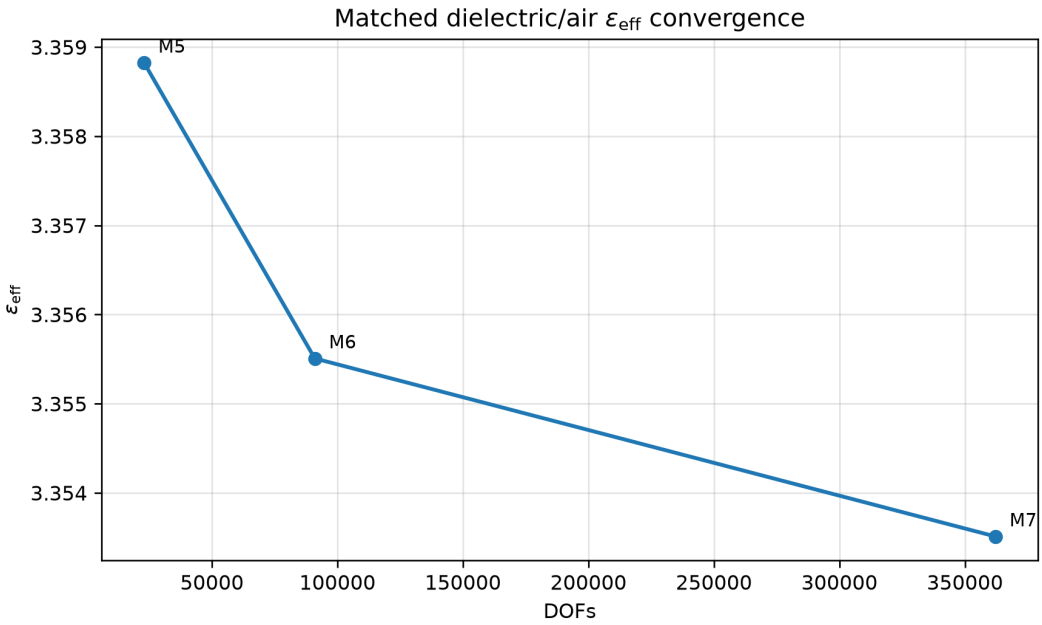


Figure 13: Matched M5–M7 effective-permittivity convergence. The M6–M7 relative change is about 0.0595%.

10 Characteristic Impedance and Engineering Validation

10.1 Final quasi-TEM impedance

Using the matched capacitance pair,

$$L = \frac{1}{c_0^2 C_{\mathrm{air}}}, \qquad Z_0 = \sqrt{\frac{L}{C}} = \frac{1}{c_0\sqrt{C C_{\mathrm{air}}}},$$

with $c_0 = 299792458\,\mathrm{m/s}$. The equivalent form

$$Z_0 = \frac{1}{c_0 C_{\mathrm{air}}\sqrt{\varepsilon_{\mathrm{eff}}}}$$

agrees to floating-point precision; this is an internal identity, not an independent validation.

Table 11: Final FEM transmission-line quantities.

Quantity	Value
C	$120.620\,\mathrm{pF/m}$
C_{air}	$35.9682\,\mathrm{pF/m}$
$\varepsilon_{\mathrm{eff}}$	3.3535
L	$3.09343 \times 10^{-7}\,\mathrm{H/m}$
v_p	$1.63708 \times 10^{8}\,\mathrm{m/s}$
Z_0	$50.642\,\Omega$

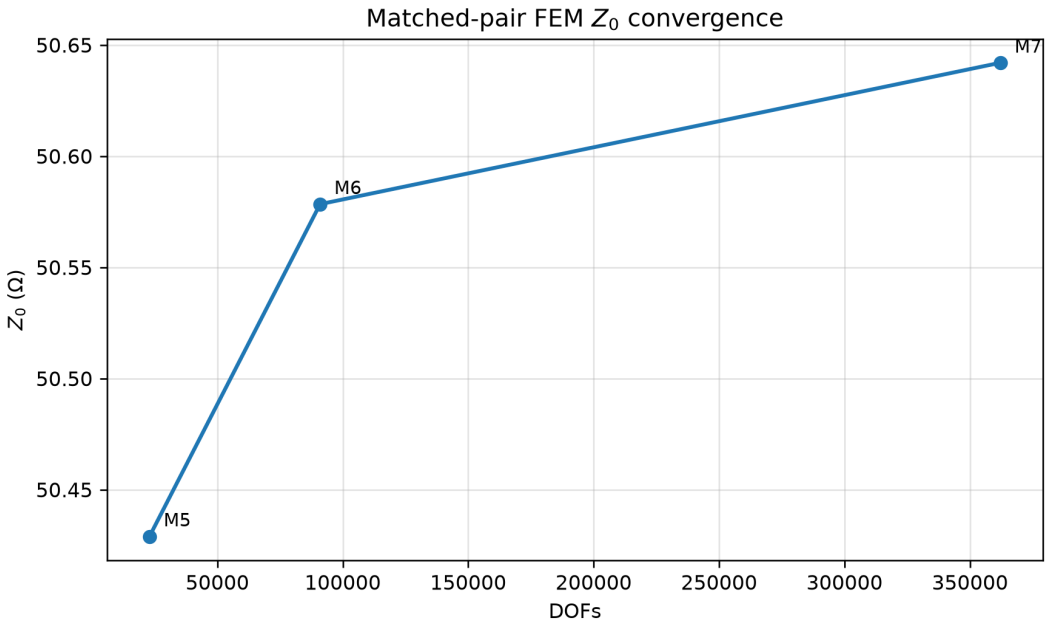


Figure 14: Direct matched-pair convergence of the final engineering quantity. The M6–M7 relative change in Z_0 is about 0.126%.

10.2 Independent engineering benchmarks

The final transmission-line table reports computed M7 outputs, with the 0.156% spatial and 0.182% finite-domain sensitivity caveats stated above; the printed digits do not establish an exact limiting value.

The Hammerstad–Jensen and IPC-2141 calculations use only geometry and ε_r, not FEM capacitance or FEM ε_{eff}. They are independent approximate engineering formulas, not exact solutions. The Hammerstad–Jensen implementation uses the finite strip-thickness correction from the project specification [1, 3]; IPC-2141 uses the compact external-microstrip expression with a natural logarithm [2, 3].

Following Appendix E of the specification, the named discrepancy uses Hammerstad–Jensen in the denominator:

$$d_{\text{HJ}}(Z) = \frac{|Z - Z_{0,\text{HJ}}|}{|Z_{0,\text{HJ}}|}.$$

The primary comparisons are therefore $100 d_{\text{HJ}}(Z_{0,\text{FEM}}) = 0.879\%$ and $100 d_{\text{HJ}}(Z_{0,\text{IPC}}) = 1.608\%$. The optional FEM-denominator differences are retained in a separately labelled column so that the two definitions cannot be confused.

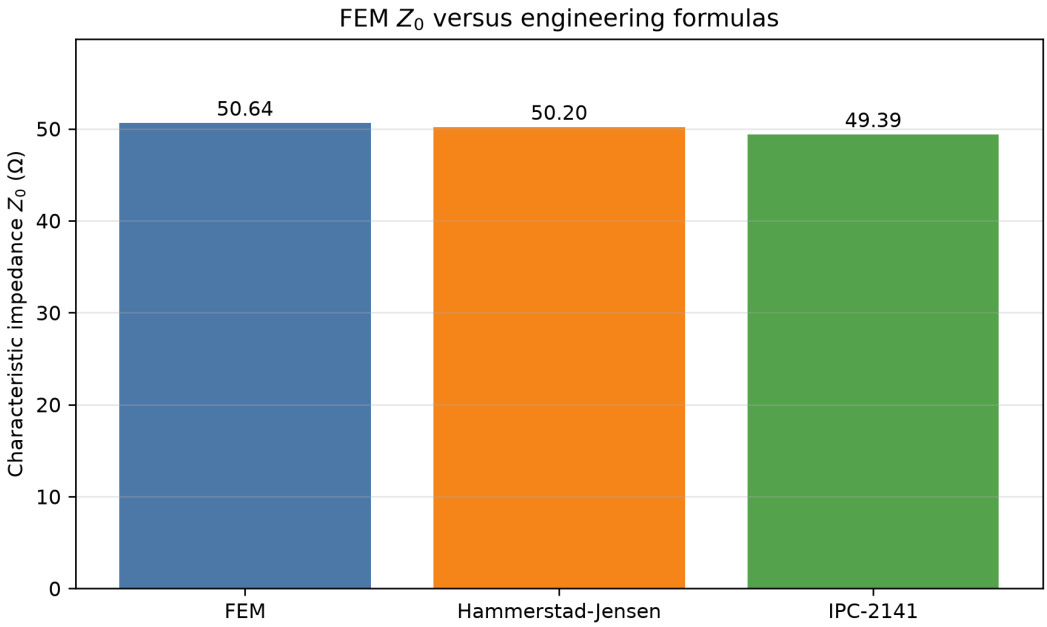


Figure 15: Final FEM characteristic impedance compared with independent engineering approximations.

Table 12: Engineering benchmark comparison. The primary d_{HJ} column uses the specification's Hammerstad–Jensen denominator. The last column is the supplementary $|Z - Z_{0,\mathrm{FEM}}|/|Z_{0,\mathrm{FEM}}|$ percentage, not d_{HJ}.

Method	$\varepsilon_{\mathrm{eff}}$	Z_0 (Ω)	$100d_{\mathrm{HJ}}$	Diff. vs FEM
FEM matched pair	3.3535	50.642	0.879%	0.000%
Hammerstad–Jensen	3.3008	50.201	0.000%	0.871%
IPC-2141	–	49.393	1.608%	2.466%

The Hammerstad–Jensen formula agrees more closely with the FEM result than the simpler IPC-2141 expression. This is useful engineering validation of the scale of the result, but the benchmark agreement does not prove FEM correctness. The numerical verification and convergence evidence above remain the primary validation.

The final FEM value is also close to the common $50\,\Omega$ design target:

$$Z_0 - 50\,\Omega = 0.642\,\Omega,$$

or about 1.284% above $50\,\Omega$. The target is design context only, not a numerical reference solution.

11 Limitations

The physical model is electrostatic and quasistatic/quasi-TEM. It does not include dielectric dispersion, dielectric loss, conductor loss, surface roughness, solder-mask loading, radiation, anisotropy, or manufacturing tolerances. The trace and ground are perfect conductors.

The numerical result also has limitations. The $8H$ domain is justified by a finite $12H$ comparison, not by an exact infinite-domain solution. The final M7 mesh was accepted through a resource-aware asymptotic fallback rather than by directly reaching the preferred 0.1% capacitance-change target. Richardson extrapolation is diagnostic only. The historical graded curves use approximate cross-mesh quadrature; only the separately recomputed exact-overlay extension is used for current quantitative efficiency claims. The adaptive comparison uses one residual indicator, one primary Dörfler parameter ($\theta = 0.5$), one symmetry/conformity closure and two declared starting meshes. It does not establish optimality among estimators or marking strategies; the indicator-to-error ratio has not stabilised.

12 Conclusions

This project produced a validated finite-element calculation of the quasistatic microstrip impedance. The final dielectric capacitance is

$$C = 120.620\,\mathrm{pF/m},$$

and the matched all-air capacitance is

$$C_{\mathrm{air}} = 35.9682\,\mathrm{pF/m}.$$

Their ratio gives $\varepsilon_{\mathrm{eff}} = 3.3535$, and the final quasi-TEM characteristic impedance is

$$\boxed{Z_0 = 50.642\,\Omega.}$$

The numerical-analysis evidence is as important as the final number. The solver passed known-solution verification with the expected P_1 rates. The domain study supported the $8H$ final domain, and the uniform energy-norm fit gave about 0.623, consistent with conductor-corner and interface regularity loss. The historical graded estimates favoured uniform refinement; the separately stored exact-overlay comparison preserves that conclusion at the formal targets. The final capacitance was selected from the uniform M3–M7 sequence, then reused exactly in a matched all-air solve. Engineering approximations gave $50.201\,\Omega$ by Hammerstad–Jensen and $49.393\,\Omega$ by IPC-2141, providing independent approximate cross-checks.

Under the tested matched-work conditions, residual-driven refinement reduced the DOFs and NNZ needed for a 0.30 relative energy error, but its cumulative algorithm time exceeded uniform refinement; improved localisation did not imply a universal runtime advantage. This extension did not replace the final computed M7 capacitance or any matched-air/impedance result.

References

[1] E. Hammerstad and O. Jensen. Accurate models for microstrip computer-aided design. In *1980 IEEE MTT-S International Microwave Symposium Digest*, pages 407–409, 1980.

[2] IPC. IPC-2141: Controlled impedance circuit boards and high speed logic design. IPC design guidance, 1996. Cited by the course technical specification as a coarse microstrip cross-check; superseded by IPC-2141A.

[3] MTH321 Teaching Team. Capacitor2d: Microstrip transmission line project specification. MTH321 Numerical PDEs coursework handout, 2026. Course-provided technical specification.

[4] MTH321 Teaching Team. Project 2 brief. MTH321 Numerical PDEs coursework handout, 2026. Course-provided project brief.

[5] MTH321 Teaching Team. Project 2 problem pack. MTH321 Numerical PDEs coursework handout, 2026. Course-provided problem pack.

[6] David M. Pozar. *Microwave Engineering*. Wiley, 4 edition, 2011. Transmission-line and microstrip background.

A AI Transparency Audit

AI/Codex assistance was used for code scaffolding, plotting routines, numerical workflow generation, documentation organisation, consistency checks, and this first report draft. The numerical results were not accepted directly from AI output. They were checked through manufactured-solution convergence, residual and matrix diagnostics, charge-balance checks, domain truncation, reference-refinement stability, matched all-air mesh identity, independent engineering formulas, and automated tests.

The audit also follows the five-gate process required for AI-assisted solver work. First, formula and equation transcription was checked against the project source documents. Second, structure

and indexing risks were tested through geometry, material-region, boundary-node, and dof checks. Third, the mathematical formulation was checked by deriving the weak form, Dirichlet elimination, capacitance identities, and quasi-TEM identities explicitly. Fourth, independent reproduction was addressed by the manufactured-solution test, mesh-refinement studies, matched air-only replay, and engineering benchmark formulas. Fifth, suspicious or fragile results were interrogated through the dedicated domain, reference-stability, graded-refinement, final-capacitance, and matched-mesh studies rather than accepted at face value.

Table 13: Project-specific AI audit summary.

Component	Main risk	Verification evidence
P1 FEM solver and Dirichlet elimination	Incorrect assembly, boundary treatment, or material coefficient handling	Manufactured-solution convergence; matrix symmetry and residual checks; material-tag tests
Microstrip geometry	Trace hole, interface, or boundary nodes misidentified	Geometry tests; full and near-trace figures; potential bounds and symmetry
Capacitance extraction	Reaction sign or trace-dof summation error	C_E/C_Q agreement and charge-balance diagnostics
Domain study	Changing the far domain could also change the near-trace mesh	Automated near-field equivalence checks and domain-truncation tables
Uniform convergence	Invalid coefficient-vector subtraction across meshes	Reference-mesh gradient transfer and reference-stability table
Graded refinement	Confusing coordinate nesting with triangular common refinement	Geometry audit detects crossed diagonals; original efficiency estimates retained with an explicit integration caveat
Final capacitance	Selecting a convenient mesh rather than an evidenced one	M3–M7 monotonic sequence, M8 preflight, Richardson diagnostic, frozen mesh artifact
Matched all-air solve	Regenerating a nominally similar mesh and contaminating C/C_{air}	Frozen mesh SHA-256, same dofs, same boundary sets, same sparse pattern, dielectric replay
Engineering benchmarks	Circular validation using FEM outputs inside formulas	Benchmark function signatures accept only geometry and ε_r; unit-invariance tests

Case audit 1: energy–charge agreement. *Claim examined:* near-equality of C_E and C_Q proves spatial convergence. *Risk and falsifier:* reactions and energy come from the same stiffness matrix, so even a coarse mesh can satisfy $2W_e = V_0 Q$. *Independent evidence:* M0 has $\delta_{EQ} \approx 4.62 \times 10^{-16}$ but $C_E \approx 167.703\,\mathrm{pF/m}$, whereas the computed M7 result is $120.620\,\mathrm{pF/m}$. *Decision:* reject that inference; use δ_{EQ} for algebraic consistency and the separate mesh sequence for spatial sensitivity. *Remaining limitation:* the mesh sequence still has the disclosed M7 fallback sensitivity.

Case audit 2: a finite far-boundary reference. *Claim examined:* the tested $12H$ solution is the exact infinite-domain answer. *Risk and falsifier:* a finite Neumann boundary still changes the field; one large box alone cannot remove truncation. *Independent evidence:* the fixed-core 3H–12H sequence and the finer 12H repeat separate measured domain and mesh effects. *Decision:* call 12H a tested finite-domain reference and report the 0.182% 8H difference separately. *Remaining limitation:* no exact infinite-domain error bound is available. This is an audited inference, not a claim that such an exact-reference error was previously committed.

Case audit 3: errors across different meshes. *Claim examined:* coefficient vectors or nested coordinate lists alone define a valid cross-mesh energy error. *Risk and falsifier:* vector entries live in different bases, and a fine triangle crossing a coarse gradient jump invalidates constant-gradient element integration. *Independent evidence:* uniform triangle-containment checks pass, while the new vertex-side test finds crossings in all five graded cases despite coordinate-subset tests passing. *Decision at the rubric review:* retain the uniform norm calculation, correct the graded guard, and withdraw precise graded efficiency claims until overlay integration is validated. *Extension verification:* Section 8 subsequently supplies separately stored exact-overlay errors for all three methods. Opposite-diagonal analytic integration, the nested Galerkin identity, material matching and area-coverage tests support that implementation. Historical curves were not overwritten. *Remaining limitation:* fine graded points remain reference-sensitive, and the legacy full-run entry point still needs a new integrated certification; the standalone extension is not presented as that certification.

Case audit 4: symmetry and mesh-quality assumptions. *Claim examined:* symmetric node coordinates alone ensure a symmetric continuous P_1 field after local refinement. *Falsifier and evidence:* historical parallel diagonals gave an approximately 2.05% continuous reflection discrepancy on M2 despite near-roundoff nodal symmetry. A first diagonal-only completion then failed coarse-companion potential bounds. *Decision:* retain the failed attempt, use a nonobtuse mirror completion of the same parent family, and test continuous reflection, bounds, conformity and material tags independently. Corrected paths pass; this is a validation-driven correction, not tuning θ to win.

Case audit 5: estimator interpretation. *Claim examined:* a residual indicator is automatically a calibrated true-error estimate. *Falsifier and evidence:* the primary reference error decreases, while the final indicator rises and the scaled indicator/error ratio changes from about 5.61 to 10.0. *Decision:* use η for marking only; report its dimensional and scaled diagnostics without a guaranteed-bound or stable-effectivity claim. Efficiency claims use independently integrated reference errors, not η.

B Individual Contribution Statements

Group C's six-member team has instructor approval, as confirmed by the team. The member identities below are supplied by the team; each member will complete their own contribution statement before submission.

Member 1:
Chuqiao Tang (2360692). **[TO BE COMPLETED BY TEAM.]**

Member 2:
Jinyu Ma (2363433). **[TO BE COMPLETED BY TEAM.]**

Member 3:
Muqing Xu (2362900). **[TO BE COMPLETED BY TEAM.]**

Member 4:
Shuoran Fu (2362626). **[TO BE COMPLETED BY TEAM.]**

Member 5:
Taiqi Zhang (2363684). **[TO BE COMPLETED BY TEAM.]**

Member 6:
Yujia Ding (2362094). **[TO BE COMPLETED BY TEAM.]**